scratch paper Name: Gabriella Ochoa ID: 1035210

9) 2 2 2
2 2 2
4 2 2
4 2
4 2

Cl - 70.6

O(4) 6(4)
= 31

5) $CaCl_2 \cdot 2H_2O$
(110.17)
Ca - 40.08
Cl(2) 35.45 = 70.09
$2H_2O(2)(18.016) = 36.032$
146.202

$\frac{110.17}{146.202} \times 100$
= 75%

6) $q = mc\Delta T$

11) $T_f - T_i = \Delta T$
$10^\circ C - 30^\circ C = -20^\circ C$

$C = \frac{q}{m\Delta T}$ $C = \frac{-80.0J}{10.0g(-20^\circ C)}$
$= \frac{80.0J}{-200}$
$= -0.400\ J/(g^\circ C)$

19) Mg^{2+} PO_4^{3+}
Mg^3 PO_4^2

20) 100g of unknown molar mass
$\frac{100g}{50\ g/moles} = 2mol$ 50 g/mol

$g = 50$ 100

15) $(NH_4)_2CO_3$
9.032

N(14.01 g/mol)(2) 28.02 g
H(4.032)(2) 8.064 g
C 12.01 g/mol 12.01 g
O 16.00 g/mol(3) 48.00 g
= 96.094 g/mol

25) 1 mol = 6.022×10^{23}

scratch paper Name: Samantha Gonzalez ID: 0907899

5.) $CaCl_2 \cdot 2H_2O$

$40.08 + (35.45 \cdot 2)$ $2 \cdot (1.008 \cdot 2) + 15.999$

$110.98\ g/mol +$ $36.03\ g/mol$

$147.01\ g/mol$

$\frac{(CaCl_2)}{CaCl_2 \cdot 2H_2O} = \frac{110.98\ g/mol}{147.01\ g/mol} = 0.75491 =$ **75.49 %**

6.) $C_p = \frac{q}{m \cdot \Delta T}$ $\Delta T = 10.0^\circ C - 30.0^\circ C$

$-20^\circ C$

$C_p = \frac{-80.0\ J}{10.0\ g \cdot -20^\circ C} = \frac{-80.0\ J}{-200\ g/^\circ C} =$ **0.400 J/g°C**

9.) valence electrons

$Cl - 7$

$O_4 - 6 \times 4$ $7 + 24 + 1 = 32$

overall charge (-)

15.) $(NH_4)_2CO_3$

$(2 \cdot 14.01\ g/mol) + (8 \cdot 1.008\ g/mol) + 12.01\ g/mol + (3 \cdot 16.00\ g/mol)$

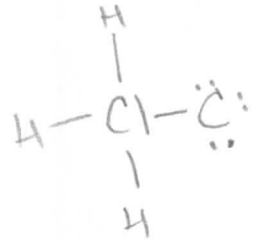
$28.02 + 8.064 + 12.01 + 48$

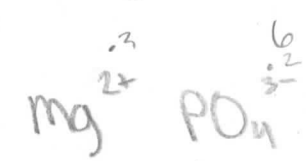
96.094 g/mol

20.)

$100g \cdot \frac{1\ mol}{50g} = 2\ mol$

25.) $6.022 \cdot 10^{23}\ atoms\ Na \cdot \frac{1\ mol}{6.022 \cdot 10^{23}\ atoms} = 1\ mol$





$Mg_3(PO_4)_2$

scratch paper Name: Vio Tavera ID: 1081907

5) % mass of $CaCl_2$ in $CaCl_2 \cdot 2H_2O$

Ca) 40.08 g/mol → +40.08 g/mol

Cl) 35.45 g/mol × 2 = 70.90 g/mol → 110.98 g/mol

H) 1.008 × 4 = 4.032 g/mol

O) 15.999 × 2 = 31.998 g/mol

36.030 g/mol → 110.98 g/mol + 36.030 g/mol = 147.010 g/mol

$$\frac{\text{molar mass of } CaCl_2}{\text{total molar mass of } CaCl_2 \cdot 2H_2O} = \frac{110.98 \text{ g/mol}}{147.010 \text{ g/mol}} \approx 0.75491 \times 100\% = \boxed{75.49\%} \ CaCl_2$$

6) Find Cp of Zn (10.0g)

$q = m \cdot C \cdot \Delta T$, $C = \frac{q}{m \cdot \Delta T}$

$\Delta T = 10°C - 30°C = -20°C$

$$C = \frac{-80.0 \text{ J}}{10.0 \text{ g Zn} \cdot (-20°C)} \approx \boxed{0.400 \text{ J/g}\cdot°C}$$

9) how many valence electrons in ClO_4^-?

Cl) 1 × 7 = 7

O) 4 × 6 = +24

31 + 1 = 32 ← −1 net charge

valence electrons can be identified on Periodic table, remove the 1 in the double digit # in each column to see how many V.E for each column :)!

15) Molar mass (g/mol) of $(NH_4)_2CO_3$

N) 2 × 14.007 g/mol = 28.014 g/mol

H) 8 × 1.008 g/mol = 8.064 g/mol

C) 1 × 12.011 g/mol = 12.011 g/mol

O) 3 × 15.999 g/mol = 47.997 g/mol

96.086 g/mol ≈ 96.09 g/mol

20) find moles of an element

given = 100g of element

given = molar mass 50 g/mol

formula:

moles = mass/molar mass

$$\text{moles} = 100 \text{ g element} \times \frac{1 \text{ mol}}{50 \text{ g/mol}} = 2 \text{ mol}$$

25) find moles

given: atoms

formula:

Moles = particles / 6.022×10^{23}

$$\text{moles} = \frac{6.022 \times 10^{23}}{6.022 \times 10^{23}} = 1 \text{ mol}$$

:)

scratch paper Name: Ariana Sutton ID: 1011137

#7. $CaCl_2 \cdot 2H_2O$

Ca: 40.08
Cl: 35.45(2) → 110.98
H: 1.008(2)
O: 15.999 → 18.015

$\% \frac{110.98}{128.995} \times 100 = 86.03\%$

#8. $6.022 \times 10^{23} = 1$ mol (Avogadro's number)

#9. $C = \frac{q}{m \cdot \Delta T}$ $C = \frac{-80.0\,J}{10.0g \cdot (-20.0)} = 0.400\,J/(g°C)$

$10.0 - 30.0 = -20.0$

#18. $14.01 + 1.008(4) = 18.042(2) = 36.084$

$12.01 + 16.00(3) = 60.01$

$36.084 + 60.01 = 96.094 = 96.09\,g/mol$

#19. $\frac{50\,g/mol}{100\,g} = 0.5\,mol$

scratch paper Name: Robert D. Ruiz ID: 1075374

Q7.) mass of $CaCl_2 \cdot 2H_2O$

$Ca = 40.08$ g/mol $H_2 = 2.016$ g/mol $2H_2O = 36.03$ g/mol

$Cl_2 = 70.90$ g/mol $O = 15.999$ g/mol $CaCl_2 = 110.98$ g/mol

$CaCl_2 \cdot 2H_2O = 147.01$ g/mol

% mass only $CaCl_2 = \frac{CaCl_2}{CaCl_2 \cdot 2H_2O} = \frac{110.98 \text{ g/mol}}{147.01 \text{ g/mol}} = 75.49\%$ when adjusted for SF

Q9.) Heat capacity of zinc

~~mq/ΔT ?~~

(mass)·(Joules)÷(Δtemp) $(10_g \text{ zinc}) \cdot (-80.0 \text{ J}) \div (-20.0°\text{C}) = 40 \text{ J/(g°C)}$

I could not remember equasion

Q1.) Cl has 7 valance electrons

O has 6 valance electrons

O_4 has 24 valance electrons

ClO_4 has 31 electrons

ClO_4^- gains 1 more, and has 32 electrons

Q8.) 1 mol = $6.022 \cdot 10^{23}$ particles. Avegadro's number

Q18.) $(NH_4)_2CO_3$ molar mass

$((14.01 + (1.008 \cdot 4)) \cdot 2) + (12.01 + (16.00 \cdot 3))$

(g/mol) (g/mol) (g/mol) (g/mol)

$(14.01 \text{ g/mol} + 4.032 \text{ g/mol}) \cdot 2$ $12.01 \text{ g/mol} + 48.00 \text{ g/mol}$

$36.084 \text{ g/mol} + 60.01 \text{ g/mol} = 96.09 \text{ g/mol}$

Q19.) 100_g at 50 g/mol

$\frac{100 \text{ g}}{1} \cdot \frac{1}{50 \text{ g/mol}} = 2 \text{ mol}$

scratch paper Name: Allen Benitez ID: 1049064

(7)

Ca : 40.08 g/mol

Cl : 35.45 g/mol (2) = 70.9 g/mol

$CaCl_2$: 40.08 g/mol + 70.9 g/mol = 110.98 g/mol

H_2O : 15.999 g/mol + 2.016 g/mol = 18.015 g/mol

H : 1.008 g/mol (2) = 2.016

O : 15.999 g/mol

$2(H_2O) = 2(18.015\ g/mol) = 36.03\ g/mol$

$$\frac{(CaCl_2)}{(CaCl_2 + 2H_2O)} = \frac{(110.98\ g/mol)}{(110.98\ \frac{g}{mol} + 36.03\ \frac{g}{mol})} \times 100 = \boxed{75.49\ \%}$$

(9) $q = mC\Delta T$

$-80.0\ J = (10.0\ g)\ C\ (-20.0\ ^\circ C)$

$\frac{-80.0\ J}{-200\ g\ ^\circ C} = \frac{-200\ g\ ^\circ C\ (C)}{-200\ g\ ^\circ C}$

$C = \boxed{0.400\ J/g\ ^\circ C}$

(19) $\frac{100\ g \mid 1\ mol}{50\ g/mol} = \boxed{2\ mol}$

(31) $\frac{1\ g\ Na \mid 1\ mol\ Na \mid 6.022 \times 10^{23}\ atoms}{22.99\ g\ Na \mid 1\ mol} = 2.62 \times 10^{22}\ atoms$

$\frac{1\ g\ Cl \mid 1\ mol\ Cl \mid 6.022 \times 10^{23}\ atoms}{35.45\ g\ Cl \mid 1\ mol\ Cl} = 1.70 \times 10^{22}\ atoms$

(18)

N : 14.007 g/mol (2) =

H : 1.008 g/mol (8) =

C : 12.011 g/mol (1) =

O : 15.999 g/mol (3) =

(35) $\frac{1.0\ mol\ SO_2 \mid 6.022 \times 10^{23}\ molecules}{1\ mol\ SO_2} = 6.022 \times 10^{23}\ molecules\ SO_2$

$\frac{0.50\ mol\ CO_2 \mid 6.022 \times 10^{23}\ molecules}{1\ mol\ CO_2} = 3.011 \times 10^{23}\ molecules\ CO_2$

$14.007\ \frac{g}{mol}(2) + 1.008\ \frac{g}{mol}(8) + 12.011\ \frac{g}{mol}(1) + 15.999\ \frac{g}{mol}(3)$

$= 96.096\ g/mol$

$\boxed{96.09\ g/mol}$

scratch paper Name: Rosa M Contreras ID: 09152336

Q5 $CaCl_2 \cdot 2H_2O$ 92.995

Ca 40.08 →

$Cl_2 \times 35.45$ 70.9

H_2 1.008 = 2.016

O 15.999 →

92.995 → 0.92995

9299.5

14098

0.837943374 x 100 =

83.79%

a) 85.98%

Q21

C

H

Cl

Na

O

Q24 d.

Q6 10.0g piece Zinc Zn cooled 30.0°C - 10.°C energy tran -80.0J

0.4000 J g°C or 2.50 J/g°C $1 kg \cdot m^2 \cdot s^2$

Q8 CO_3^{2-}

Hill system

C

O

Q9, 10, 11, 12

Q9 Cl 7 +

0.6 x 4 = 24 + 7 31

Q25

Na 58.69

6.022×10^{23}

9.745 9

9.746×10^{23}

H_2O

Q13



O 6

H 1 x 2 = 2

Trigonal planar

Q14

Na 5

Cl 7

12

ionic

~~nonpolar covalent~~

polar covalent

~~metallic~~

Q15

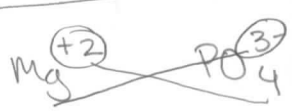
N_2 x 14.01 g/mol 28.02

H_6 x 1.008 g/mol 6.048

C_1 x 12.01 g/mol 12.01

~~O_3 x 16.00 g/mol~~ 48 94.078

Q19



$Mg(PO_4)_2$

scratch paper Name: Carina Mendoza ID: 0718768

⑤ $CaCl_2 \cdot 2H_2O$

Ca = 40.08

Cl = 35.45

O = 16.00

H = 1.008

$CaCl_2$ = $40.08 + 35.45 + 2(16.00) = 107.53$

$2H_2O = 2[2(1.008) + 16.00) = 36.032$

$$\frac{107.53}{107.53 + 36.032} \times 100 = 74.90$$

143.542

~~Sodium = 22.99~~

⑥ $30 - 10 = 20$

$$\frac{-80}{10.0(20.0)} = -0.400 \text{ J}$$

15. What is the molar mass (units = g/mol) of ammonium carbonate, $(NH_4)_2CO_3$

144.18
+ 112.09
96.09
72.05

43.028 ÷ 4 = 10.757

scratch paper Name: Nhia Lee ID: 0884241

Q6. ClO_4^-

$7 + (6 \times 4) + 1$

$= 32$

Q8. 1 mol $= 6.022 \times 10^{23}$

Q9.

mass - 10.0 g

heat - 20.0°c

Intial

final

Q7. $CaCl_{(2)} \cdot 2H_2O$

$Ca = 40.08$ $2 \times 18.016 = 36.032$

$Cl = 35.45 \times 2$

$40.08 + (35.45 \times 2)$

$110.98 + 36.032 =$

147.012

$\frac{110.98}{147.012} \times 100 = 0.75.49$

75.49%

30.0°c - 10.0°c

$C = \frac{q}{m \cdot \Delta T}$

Q18 $(NH_4)_2CO_3$

$14.01 + 14.01 = 28.02$

$4.032 + 4.032 = 8064$

12.01

$16.00 \times 3 = 48$

$28.014 + 8.064 + 12.01 + 48$

28.02 12.01

$= 96.094$

Q19 $\frac{100\ g}{50\ g/mol} = 2\ mol$

scratch paper Name: Fernanda Dominguez ID: 0981154

$CaCl_2 \cdot 2H_2O$

Q5: Ca: $1 \times 40.08 = 40.08$ > 110.98
Cl: $2 \times 35.45 = 70.9$

H: $2 \times 1.008 = 2.016$ > 18.016
O: $1 \times 16.00 = 16.00$

$110.98 + 18.016 = 128.996$

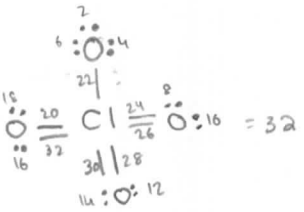
$\frac{110.98}{128.996} \times 100\% \approx 86.0336 \approx 85.98$

Q6: $q = m \cdot c \cdot \Delta T$

$-80.0j = 10.0g \cdot \quad \cdot 20$

$= .200j$

Q9:



Q15:

Q20:

$\text{Mole} = \frac{\text{mass (g)}}{\text{molar mass(g/mole}}$

$\text{mole} = \frac{100}{50 g/mo}$

$= 2 \text{ mole}$

Q25:

$6.022 \times 10^{23} = 1 \text{ mole}$

$\text{mole} \times \frac{6.022}{1} = _____$

$1 \times 6.022 \times 10^{23} = 6.022 \times 10^{23}$

scratch paper Name: Angel Sanchez ID: 0848677

Q.7 $Ca = 40.08$, $Cl = 35.45\ g/mol \times 2$, $H = 1.008 \times 4$, $O = 16.00 \times 2$

$110.98\ g/mol / 147.012\ g/mol$

$\approx 75.50\%$

Q.9 $\Delta T = T_F - T_I$

$C_p = \frac{q}{m \times \Delta T} = \frac{-80.0\ J}{10.0\ g \times -20^\circ C} = \frac{-80\ J}{-200\ g^\circ C} = .400\ J/g^\circ C$

Q.18 Molar mass g/mol $(NH_4)_2CO_3$

$N = 14.01\ g/mol \times 2 =$

$H = 1.008 \times 8 =$

$C = 12.01\ g/mol =$

$O = 16.00 \times 3 =$

Molar mass = 96.094

$\approx 96.099\ g/mol$

Q.19 moles = $100\ g / 50\ g/mol = 2$ moles

Q.10 ClO_4^- = Gilchrist

Cl = Gilchrist $= 7$

$O_4^- = 6 \times 4 = 24$, add one electron for each − charge: $24 + 1 = 25$

$25 + 7 = 32$

Q.8

Moles = 6.022×10^{23} atoms $\div$ 6.022×10^{23} (avogadro's number)

$= \boxed{1\ mol}$

scratch paper Name: Carlos Bedolla ID: 1058710

1.0 865 0722 e25

8.4369621 e 24
2.4281107 2 e24 2
7.23.25421 e24
2.8906 08 e25

27 -  linear

28-  0=0

29 - vaporization/boil ↳ endothermic ✓

33 - H-Bond > dipole-dipole > LDF ✓

scratch paper Name: Cynthia Sanchez ID: 1093672

18

18.012

14.01 g/mol

1.008 g/mol x4 4.032 x2

12.01 g/mol 12.01

16.00 g/mol x3 48

48
+ 12.01
36.084

96.094

23

Mg^{2+} PO_4^{3-}

$Mg_3(PO_4)_2$

scratch paper Name: Emily M ID: 0822239

$CaCl_2$ % of Mass

$Ca(1 \times 40.08) = 40.08$

$Cl\ (2 \times 35.45) = 70.09$

$= 110.98$ g/mol

$\frac{40.08}{110.98} \times 100 = 36.11$

$\frac{70.09}{110.98} \times 100 = 63.15$

$63.15 - 36.11 = 27.04$

$(NH_4)_2CO_3$

Molar Mass

$N = 14.01$ g/mol

$H = 1.008$

$C = 12.01$ g/mol

$O = 16.00$

$(4 \times 14.01) = 56.04$

$(4 \times 1.008) = 4.032 \times 2 = 60.072 \times 2 = 120.144$

$(1 \times 12.01) = 12.01 = 60.01$

$(3 \times 16.00) = 48$

$\frac{120.14}{60.01}$

20) $\frac{60g}{50g/mol}$ how many mol =

Mole $= \frac{50g/mol}{100g/m} =$

29) 60.22×10^{23} atoms Na

how many moles of Na = Avogadro #

scratch paper Name: Lizett Castro ID: 991318

7) 6.022×10^{23} g Sodium (Na)

9) 10.0g (Zn) 30.0°C to 10.0°C Energy -80.0J

10.0g (Zn) - 30.0°C +10.0°C

-20.0°C -80.0J

18)

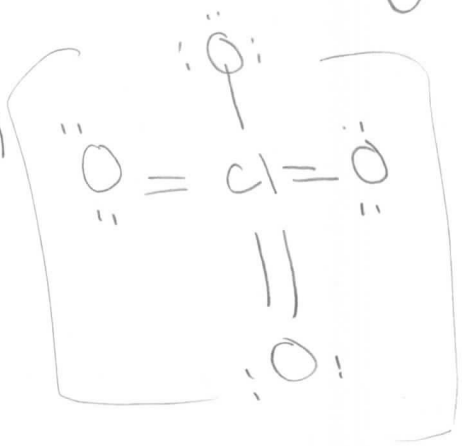
N = 14.01 g/mol H = 1.008 g/mol C = 12.01 g/mol O = 16.00 g/mol

14.01/1.008/12.01/16.00 = 0.0723293558

19) $\frac{100}{50}$ = 2 moles

100g ÷ 50 g/mol

11)



a) ClO_4^- = ~~35.45 + (16.00 x 4)~~ Cl = 7 ; O = 6 x 4 ; $x^- = 1 = 31 + 1 = 32$ valency

scratch paper Name: ID:

$H_2O = 18.02$

Q7: $= CaCl_2 \cdot 2H_2O$ Wet water

$40.08 + 70.9 \cdot 36.04$

Ca Cl2

$\frac{110.98}{147.02} = 75.49$

Q8:

$moles = \frac{6.022 \times 10^{23}}{1} = 1 mol$

Q9: $C = \frac{q}{m \cdot \Delta T} = \frac{-80.0\,J}{10.0g \cdot 30-10} = \frac{-80.0\,J}{10.0g \cdot 20} = -0.4\,J/(g°C)$

Q: 18 $= (NH_4)_2CO_3$

$2NH_4CO_3$ (N x2, H x8)

$14.01 + 1.008 + 12.01 + 16 \times 3$

$\downarrow$

$28.02 + 9.064 + 12.01 + 48 = 96.094$

Q: 19 =

$moles = \frac{mass}{molar\ mass} = \frac{100}{50} = 2\ mol$

Name: Jose Sanchez

Student Id: 1035463

scratch paper Name: Aaron Robles ID: 0955892

7. $CaCl_2 \cdot 2H_2O$

$H = 1.008 \times 4 = 4.032$ g/mol H

$O = 16 \times 2 = 32$ g/mol O

$Ca = 40.08$ g/mol Ca

$Cl = 35.45 \times 2 = 70.9$ g/mol Cl

$= 110.98$ g/mol $CaCl_2$

$32 + 4.032 + 40.08 + 70.9$

147.012

$\frac{110.98}{147.012} = 0.7549 \times 100 = 75.49\%$ of $CaCl_2$

8.

6.022×10^{23} $Na = 23$

18.

$N = 14.01 \times 2 = 28.02$

$H = 1.008 \times 8 = 8.064$

$C = 12.01$

$O = 16 \times 3 = 48$

$28.02 + 8.064 + 12.01 + 48. = 96.094$ molar mass

23.

Mg^{2+} PO_4^{3-} $Mg_3(PO_4)_2$

scratch paper Name: Alyssa Martinez ID: 0931469

18.) $14.01 \overset{?}{g/mol} \times 1.008\ g/mol \times 12.01\ g/mol \times 16.00\ g/mol$

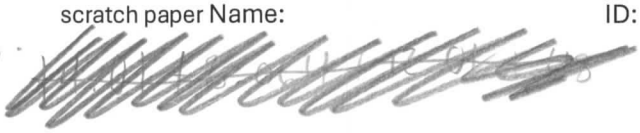
$= 144.18$ (calc. gave me complete different answer)

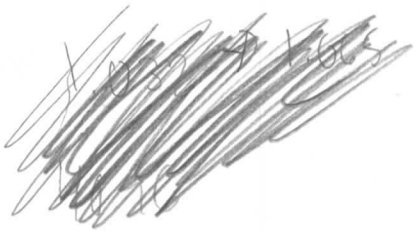
19.) $\frac{50\ g/mol}{100\ g/mol} = 0.5\ mol$

8.) 6.022×10^{23} Atoms

scratch paper Name: ID:

18.





scratch paper Name: Isaac Mack ID: 0936354

(5) $1Ca(40.08) + 2Cl(35.45) + 2H(1.008) + 4O(15.999)$

$40.08 + 70.9 + 2.016 + 63.996$

mass of salt CaCl → $110.98 + 66.012$

176.992

$\frac{110.98}{176.992} = 0.627$ or 62.7% ~~≈63.92%~~

(6.) IDK

However. ~~Logically~~, its all very clean numbers, and (g°C) makes me think multiplication 10g and 20°C change with the only other value being 80. 0400 seems likely ☺

(7.) O has 6
Cl has 7

$6 \times 4 \rightarrow 24$
$7 \times 1 \rightarrow 7$
> 31 odd??

but +1 from negative charge! (32)

(10) if you have 100g
and molar mass is 50
grams PER mol

... its 2

why must i show work for this.

(15) N 8 ← whoops there's 8 not 4

$2NH_8C_3O$

$2(14.007) + 8(1.008) + 1(12.011) + 3(15.999)$

~~92.054~~ ≈ 96.09 ☺

scratch paper Name: JUSTIN VINK ID: 1070339

Q21

2.5 - 2.1 = 0.4

3.5 - 2.1 = 1.4

0.9 - 3.0 = 2.1

0

~~Ca~~ $Mg_3(PO_4)_2$

Q9

7

6x4 = 24 +1

32

Q5

40.08

35.45 x 2 110.98

1.008 x 4 ~~4.032~~

15.999 x 2 36.03

147.01 75.49

Q15

14.01 x 2 $^{2}28.02$

1.008 x 8 + 8.064

12.01 + 12.01

16.00 x 3 + 48

96.094

Q6

-20

$\frac{-80}{\cancel{-80}} = \frac{10 \times C \times -20}{10}$

$10 \times Cx - 20 + 80$

$\frac{-8}{-20} = \frac{Cx - 20}{-20}$

$\frac{4}{10} \quad \frac{2}{10}$



Q20

100 g | 50 g/mol = 2 mol

Q25

6.022×10^{23} atoms = 1 mol

scratch paper Name: Maria Gomez ID: 0771844

Question 5: mass% $CaCl_2$ in $CaCl_2 \cdot 2H_2O$

$\frac{\text{mass element}}{\text{mass compound}} \times 100\%$

(1) $Ca = 40.08\ g/mol$

(2) $CL = 35.45\ g/mol = 70.9$

(1) $H = 1.008\ g/mol = 4.032$

(2) $O = 15.999\ g/mol = 31.998$

$40.08 + 70.9 + 4.032 + 31.998 = 147.01$

Total mass 147.01

$CaCl_2 = 46.08_g + 70.9_g = 110.98$

Double check!

$2H_2O = 4.032 + 31.998 = \frac{36.03}{147.01} = .2450 \approx 24.50\%$

$\frac{110.98}{147.01} = 0.7549 \times 100 = \boxed{75.49\%}$

Question 6:

$q = C \times m \times \Delta T$

$\Delta T = T_f - T_i$

Known:

$m = 10.0g$

$\Delta T = 10.0^\circ C - 30.0^\circ C = -20^\circ C$

$q = -80.0J$

unknown? C

$C = \frac{q}{m \cdot \Delta T}$ $C = \frac{-80.0J}{10.0g \times -20.0C}$ $C = \frac{-80.0J}{-200\ g/^\circ C}$

$\boxed{C = 0.400\ J/(g^\circ C)}$

Question 9:

ClO_4^-

Cl = 7 valence electrons

O = 6 valence electrons (x4) = 24

+1 electron for (-) charge

$7 + 24 + 1 = \boxed{32 \text{ total}}$

Question 15: molar mass $(NH_4)_2CO_3$

$N(2) = 14.007 \times 2 = 28.014$

$H(8) = 1.008 \times 8 = 8.064$

$C(1) = 12.011 \times 1 = 12.011$

$O(3) = 15.999 \times 3 = 47.997$

$96.08/6 = \boxed{96.09}$

Question 20: 100g of element

molar mass = 50g/mol

moles?

Known:

$m = 100g$

$mm = 50g$

unknown: moles

$\text{moles} = \frac{\text{grams}}{\text{molar mass}}$

$\frac{100g}{50g} = 2\ mol$

Question 25:

6.022×10^{23} atoms Na → moles?

$6.022 \times 10^{23}\ \text{atoms Na} \times \frac{1\ mol}{6.022 \times 10^{23}} = \boxed{1}$

scratch paper Name: Emily Galicia ID: 1028935

Q1. $6 \times O_4 = 24$

$7 \times Cl = 7$

31

31 electrons

Q8. 6.022×10^{23} Sodium (Na

How many moles?

22.99: molar mass

Q19. $\frac{100g}{50g/mol}$

2mol

Q7. $CaCl_2$

Ca = 1 MM = 40.08

Cl = 2 Molar Mass = 35.45

$35.45 \cdot 2 = 70.90$

110.98

H: $1.008 \cdot 4 = 4.032$

2: $15.999 \cdot 2 =$ 31.998

36.03

Q18. $(NH_4)_2CO_3$

$N = 14.007 \times 2 = 28.014$

$O = 15.999 \times 3 = 47.997$

$C = 12.011$

$H = 1.008 \times 8 = 8.064$

Molar Mass = 96.086 g/mol

= 96.09 g/mol

Q9.

10.0g, 30°C - 10°C, -80.0J

Q35.

$\frac{1.0}{64.058}$ $\frac{0.50}{44.009}$

1

Q31. $\frac{1.00g}{22.99}$ $\frac{1.00g}{35.45}$

0.043497173 0.028208745

1.699×10^{23}

2.619399

2.62×10^{23}

scratch paper Name: Mandy Yang ID: 0869542

5. Finding: % $CaCl_2$

$$\%\ CaCl_2 = \frac{75.53}{110.98} \times 100$$

$$= 63.92\%\ CaCl_2$$

6. Finding c (specific heat) J/(g°C) of zinc

$\Delta T = Tf - Ti \rightarrow 10.0°C - 30.0°C = -20.0°C$

$c = q/(m\,\Delta T)$

$c = -80.0\,J/(10.0 \cdot -20.0°C)$

$= 0.400\ J/(g°C)$

15. Molar mass of $(NH_4)_2CO_3$

$N = 14.007 \times 2 = 28.014$

$H = 1.008 \times 4 = 4.032 \times 2 = 8.064$

$C = 12.011 \times 1 = 12.011$

$O = 16.00 \times 3 = 48.00$

$28.014 + 8.064 + 12.011 + 48.00$

$= 96.089\ g/mol$

9. Cl = 7 valence e^-.
O = 6 valence $e^- \rightarrow 6 \times 4 = 24$
$24 + 7 = 31$ valence e^-

25. $1\ mol = 6.022 \times 10^{23}$

20.
Given: 100 g
50 g/mol
Find: mol
$100/_{50} = 2$ mol

21. H = 2.1 C = 2.5 O = 3.5 Na = 0.9 Cl = 3.0
Polar covalent?
C-H = 2.5 - 2.1 = 0.4
O-H = 3.5 - 2.1 = 1.4 ✓
Na-Cl = 3.0 - 0.9 = 2.1
C-C = 2.5 - 2.5 = 0

scratch paper Name: Galilea Leon ID: 09041104

5) $CaCl_2 \cdot 2H_2O$ divide w/ total mass

Ca 40.08

Cl 35.45

$40.08 + 35.45(2) = 110.98$ + $= 147.012$

$2+H2O = H2O = 18.016\ (2) = 36.032$

$\frac{110.98}{147.012} \times 100 = 75.5$

6) $q = mC\Delta T$ solve for C Δt F-I

$-80.0\ J = 10.0\ g\ C\ (10.0 - 30.0^\circ C)$

$\frac{-80.0\ J}{10.0\ g \cdot -20.0} = \frac{10.0\ g\ C\ (-20.0^\circ C)}{10.0\ g \cdot -20.0}$ $0.4 = C$

15) $(NH_4)_2CO_3$

$(N + 4(H1)) \times 2 + C + 3(O)$

$(14.01 + 4.032)2 = 36.084 + 12.01 + 48 = 96.094$

20) $100g = \frac{50g}{1mol} =$

$\frac{100g}{50g}$

grams → moles = divide by molar mass

$100g \div 50 = 2mol$

25) $6.022 \times 10^{23} \div 6.022 \times 10^{23} = 1mol$

avogradros # — Na

scratch paper Name: Sydell Lyon ID: 1111885

(7) $Ca = 40.08$

$Cl_2 = 35.45 = 70.9$

73.53

(8) ~~(8)~~ $Na = 22.99$

(9) $10.0g$ $38.0°C - 10.0°C$

$-80.0J$

$-80 + 273.15 = 193.15$

(18)

$N = 14.01$ $+14.01$

$H = (1.008) \cdot 4 = 4.032$

$18.042 \times 2 = 36.084$

$C = 12.01$ 12.01

$O = 16.00 \times 3 = 48.01$

$60.01 \times 3 = 180.03$

216.114

30.026 18.042

$+ 60.01$

78.052

(19) $\frac{100g}{50g/mol} = 2$

scratch paper Name: Guisela Gallardo ID: 0577380

5) $Ca = 40.08$
$Cl = 35.45 \times 2 = 70.9$
$+40.08$
110.98

$H = 1.008 \times 2 = 2.016$
$O = 15.999 + 15.999$
$18.015 \times 2 = 36.03$

$110.98 + 36.03 = 147.01$

$\frac{110.98}{147.01} \times 100\% = 75.49\%$.

6) $m = 10.0\,g$ $\Delta T = -20^\circ C$ $q = -80.0\,J$
$C = \frac{q}{m \cdot \Delta T}$ $\frac{-80.0\,J}{10.0\,g \times -20^\circ C} = 0.4\,J/(g^\circ C)$

9) $Cl = 7$
$O = 6 \times 4$
$31 + 1 = 32$

15)
$N = 14.01 \times 2 = 28.02$
$H = 1.008 \times 8 = 8.064$
$C = 12.01 = 12.01$
$O = 16.00 \times 3 = +48$
96.094 mass

20) $100\,g$
$50\,g/mol$
$moles = \frac{100\,g}{50\,g/mol} = 2$ mole

25)

scratch paper Name: Charlie Hatch ID: 0992103

Q7.) 1 Ca = 40.08
2 Cl = 70.9
4 H = 4.032
2 O = 31.998

$31.998\ g\ O + 4.032\ g\ H_2 = 36.03\ g\ H_2O$

$\frac{36.03}{147.01} \times 100\% = 24.509\%\ H_2O$

$100\% - 24.51\% = 75.49\%\ CaCl_2$

Q8.) $6.022 \times 10^{23}\ \text{atoms Na} \times \frac{1\ \text{molecule Na}}{1\ \text{atom Na}} \times \frac{1\ \text{mole Na}}{6.022 \times 10^{23}\ \text{molecules}} = 1\ \text{mol Na}$

Q9.) $C_p = \frac{-80.00}{10.0\ g \times (30.0°C - 20.6°C)} = -0.8\ ??\ J/(g°C)$ not sure what I did wrong

Q18.) $(NH_4)_2CO_3$

masses
N = 28.02
H = 8.064
C = 12.01
O = 48.00

$\left(\frac{2\ \text{mol N}}{1\ \text{mol } (NH_4)_2CO_3} \times \frac{14.01\ g\ N}{1\ \text{mol N}}\right) + \left(\frac{8\ \text{mol H}}{1\ \text{mol } (NH_4)_2CO_3} \times \frac{1.008\ g\ H}{1\ \text{mol H}}\right)$

$+ \left(\frac{1\ \text{mol C}}{1\ \text{mol } (NH_4)_2CO_3} \times \frac{12.01\ g\ H}{1\ \text{mol C}}\right) + \left(\frac{3\ \text{mol O}}{1\ \text{mol } (NH_4)_2CO_3} \times \frac{16.00\ g\ O}{1\ \text{mol O}}\right)$

$= 96.09\ g/\text{mol}$

Q19.) 100 g Sample element molar mass = 50 g/mol

$\frac{100}{50} = 2\ \text{mol}$

Q23.) $Mg^{2+} + PO_4^{3-} = Mg_3(PO_4)_2$

because

$2+ \times 3 = 6+$ combined with $6- = 3- \times 2$

= neutral charge

scratch paper Name: Cecilia Kongsee ID:

7)
Ca 40.08 } 110.98
Cl_2 35.45 }
+ 147.01
2 H_2 1.008 } 36.03
O 15.0999 }

$(119.98/147.01) \times 100 = 75.49146317$

8) $1\ mol = 6.022 \times 10^{23}\ g$

9) Zn = 63.39 avg mass
subscripts?
$\left(\frac{(63.39^2/_{s^2})}{100}\right)$ 1 Joule

1) Cl = 7 elec
O = 4 elec
$4 \times 4 = 24 + 7 = 31$ electrons
O Cl

18) $(14.01 + (1.008 \cdot 4) \cdot 2) + 12.01 + (16.00 \cdot 3)$
66.95940476

19) 1 mol × 2 = 2 mol
↓
50 g × 2 = 100 g

scratch paper Name: Jennifer Ibarra ID: 0873148

Q9) ClO_4^-

Cl = 7
O = 24
31
+ 1
32

Q5 $CaCl_2 \cdot H_2O$

Ca = 40.08 O = 15.999
Cl 35.45×2 H = 1.008×2

$\frac{\text{mass of element}}{\text{total mass}} \times 100$

$\frac{110.98}{128.995} \times 100 =$ 86.034

Q25)

6.022×10^{23} atoms $\times \frac{1 \text{ mole}}{6.022 \text{ atoms}} \times \frac{22.99}{1 \text{ mole}} =$ 22.99 mol

Q20) $\frac{1}{100g} = \frac{50g}{1 \text{ mol}} \Rightarrow \frac{50}{100 \text{ mol}} =$ 0.5 mol

Q15 $(NH_4)_2CO_3$

O = 16 × 3 = 48
C = 12.01
N = 14.01 ×2
H = 1.008 × 8
96.094

scratch paper Name: Oluwapelumi Tay ID:

Question 7 (4 points)

$CaCl_2 \cdot 2H_2O$

$= 1 \times 40.08 + 2 \times 35.45 + 4 \times 1.008 + 2 \times 16$

$= 147.012$ (molar mass of $CaCl_2 \cdot 2H_2O$)

Molar mass of $CaCl_2$

$= 1 \times 40.08 + 2 \times 35.45 = 110.98$

$\therefore$ % Mass of $CaCl_2 = \frac{110.98}{147.012} = 0.7549$

$= 75.49\%$

Question 9

$q = m \times c \times \Delta T$

$-80.0J = 10.0g \times c \times (10.0^\circ C - 30.0^\circ C)$

$c = \frac{+80.0J}{10.0g \times +20.0^\circ C} = 0.400 J/(g^\circ C)$

Question 18

Molar mass of $(NH_4)_2CO_3$

$2(14.01) + 8(1.008) + 12.01 + 3(16)$

$= 96.09 g/mol$

(2 point question)

Question 1

Number of valence electrons for ClO_4^-

Oxygen has 6 electrons/dots

Total number of electrons $= 6 \times 4 = 24$

Chlorine has 7 electrons

Total valence electrons $= 24 + 7$

$= 31$ valence electrons

Question 8

Atoms per molecule = Atom × No. of molecules

Atom = Moles × Avogadro's number

$6.022 \times 10^{23} = \text{moles} \times 6.022 \times 10^{23}$

$\text{Moles} = \frac{6.022 \times 10^{23}}{6.022 \times 10^{23}} = 1 \text{ mol of Na}$

Question 19

$\text{Moles} = \frac{\text{Mass}}{\text{Molar mass}}$

$= \frac{100\text{g}}{50.0\text{g/mol}}$

$= 2 \text{ moles}$

2.5-2.1 3.5-2.1 .9-
.4 1.4

scratch paper Name: Martin Garcia ID: 0681538

Q7. $CaCl_2 \cdot 2H_2O$

$40.08 + (2 \times 35.45) + 2 \cdot (18.02)$

147.02 g/mol

$CaCl_2$

$40.08 + (2 \times 35.45)$

$40.08 + 70.9$

110.98 g/mol

Percent mass

$\frac{110.98 \text{ g/mol}}{147.02 \text{ g/mol}} \times 100\%$

$= 75.489\%$

75.49%

Q9

$q = m \times c \times \Delta T$

$-80.0 \text{ J} = 10.0 \text{ g} \times c \times (30.0 - 10.0^\circ\text{C})$

$\frac{-80.0 \text{ J}}{(10.0 \text{ g})(20.0^\circ\text{C})} = c$

$\frac{-80.0 \text{ J}}{200.0 \text{ g}^\circ\text{C}} = -0.400 \text{ J/g}^\circ\text{C}$

Q18. $(NH_4)_2CO_3$

$2(14.007) + (4 \times 1.008) + 12.011 + 3(15.999)$

$2(14.007 + 4.032) + 12.011 + 47.997$

$2(18.039) + 12.011 + 47.997$

$36.078 + 12.011 + 47.997$

96.086

96.09 g/mol

scratch paper Name: Ezabella Garcia ID: 0913233

5.) $CaCl_2 \cdot 2H_2O$

$Ca = 40.08$

$Cl = 35.45 \times 2 = 76.9$

110.98

$H = 2 \times 1.008 = 2.016$

$O = 16.00$

$2(16.00 + 2.016)$

$2 \times 18.016 = 36.03$

$\frac{110.98}{147.01} \times 100 = 75.49\%$

6.) $C = \frac{-80}{10.0 \times 20} = -0.400\ J(g^{\circ}C)$

$30 - 10 = 20$

15.) $(NH_4)_2$

$N = 14.01 \times 2 = 28.02$

$H = 1.008 \times 4 \times 2 = 8.064$

$= 36.084$

CO_3

$C = 12.01$

$O = 3 \times 16.00 = 48.00$

$= 60.01$

$= 96.09$

20.) $100g \div \frac{50g}{1\ mol} = 2\ mol$

25.)

scratch paper Name: Deianna Adams ID: 1111240

6.

$$\frac{10.0\,g}{10\,x} = \frac{30 - 3n \quad 2.50\,J/g\,^\circ C}{80}$$

+30
+10
+10

-50

scratch paper Name: mina ID: 0934994

5.) $CaCl_2 \cdot 2H_2O$ — 18.015

$CaCl_2$ — 70.9 = 53.465

40.08 | 35.45×2 = 70.9

$\frac{110.98}{128.995} = 86.$

20.) $\frac{50\,g}{100\,g/mol} = 2\,mol$

25. 6.022×10^{23} atoms

particles → mol = ÷ 6.022×10^{23}

~~B~~

$\frac{22.99}{6.022 \times 10^{23}}$

$\frac{22.99 \times 3 = 68.97}{6.022 \times 1023 = 1\,mol} = 1.145$

15.) 144.18 g

1.008

12.01

16.00

6 10 g

30°C - 10°C

20°C

6-10

scratch paper Name: Sabrina Garcia ID: 0490011

5 40.08 C
35.45 Cl $\frac{40.08}{35.45} \times 100$

6 65.39 (Zn)

15 14.01 + 1.008 + 12.01 + 16.00

9 2+2+2+1 = (7)

20

25 22.99 (Na)

$6.022 \times 10^{23} \times 22.99 =$

1.384

OR 22.99 round # 2

scratch paper Name: Layla Moreno ID: 10960633

1.) ClO_4^-

$7+(6\times4)+1=$ 32 electrons

8.) atoms → mol

6.022×10^{23} atoms Na = 1 mol Na

7.) $CaCl_2 \cdot 2H_2O$

$40.08+(35.45\times2)\cdot 2\times(1.008\times2)+15.999$

$110.98 \cdot 36.03$

147.01

$(110.98 \div 147.01)\times100 =$ 75.49%

9.) $C = \frac{q}{m(\Delta T)} \rightarrow C = \frac{-80.0\,J}{10.0\,g(30°C-10°C)} =$ -0.4 J/g°C

13.) $H_2O \rightarrow$ 

2+6=8 valence electrons

electron geometry = 4 = tetrahedral

Molecular Geometry: bent

14.) Na Cl

0.9 3.0 → 3.0 − 0.9 = 2.1 = ionic | metal ; nonmetal ✓

17.) Rb = 0.8 Cl = 3.0

3.0 − 0.8 = 2.2 = ionic ✓ | metal ; nonmetal

18.) $(NH_4)_2CO_3$

$(14.007\times2)+(1.008\times8)+12.011+(15.999\times3)$

= 96.086 g/mol

19.) mass ÷ molar mass = moles

100 g ÷ 50 g/mol = 2 mol

20.) C-H bond
2.5 − 2.1 = 0.4
nonpolar

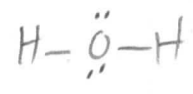
O-H bond
3.5 − 2.1 = 1.4
polar

Na-Cl
0.9 3.0 = 2.1
ionic

C-C bond
2.5 − 2.5 = 0
nonpolar

scratch paper Name: Skye Lee ID: 1073585

1 ?
~~2~~ ?
~~3~~
~~4~~
~~5~~
~~6~~
~~7~~
16 ?
~~17~~



$Mg_3(PO_4)_2$

Q5 Ca = 40.08 g/mol
Cl = 35.45 g/mol (2 times)
Hydrogen = 1.008 g/mol (2 times)
Oxygen = 15.999

2 more times

40.08
+ 35.45 (2)
110.98
110.98 g/mol
147.01 g/mol
= 0.7549
x 100
75.49 %

(1.008 x 2) + 15.999
= 18.015 g/mol
x 2
36.03
+ 110.98 g/mol
147.01 g/mol

Q6 10.0°C − 30.0°C = −20.0°C

$\frac{-80.0\,J}{(10.0\,g)(-20.0^\circ C)} = 0.400\,J/g^\circ C$

Q9 O = 6 valence electrons
Cl = 7 valence electrons
6 x 4 = 24 valence electrons
+ 7
31 valence electrons



gives us another valence electron
31
+ 1
32 valence electrons

Q15 N = 14.01
H = 1.008 (4) 2 times
C = 12.01
O = 16.00 (3)
14.01
+ 1.008 (4)
18.042
x 2
36.084
+ 12.01
+ 16.00 (3)
96.094 g/mol
= 96.09 g/mol

Q25 $\frac{160\,g}{509/mol} = 2\,mol$

6.022×10^{23} atoms Na = 1 mol

1 mol | 6.022×10^{23} | Avogadro's | atoms Na

scratch paper Name: Shane Xiong ID: 1012185

1.) $e_{Total} = (V_{electron} + Charge^{+/-})$ Oxygen Electrons = 6
Chlorine Electrons = 7
Ion Charge = 1

$= (6 \times 4) + 7 = 31 + 1 = \boxed{32}$

7.) $CaCl_2 \cdot 2H_2O$

$Ca = 40.08$
$Cl_2 = 35.45 \times 2 \} = 110.98$
$H_2 = 1.008 \times 2 \}$
$O = 15.999 \} \times 2 = 36.03 \} = 147.01$

$\frac{110.98}{147.01} \times 100 = \boxed{75.49\%}$

8.) Atoms $\rightarrow$ Moles $= \frac{\#Atoms}{Moles} = \frac{6.022 \times 10^{23}}{6.022 \times 10^{23}} = \boxed{1 \text{ mol}}$

9.) $M = g = 10.0\text{g}$ $\Delta T = 30.0^\circ C - 10.0^\circ C = 20.0^\circ C$

$q = J = 80.0\text{J}$

$C = \frac{J}{g \cdot C^\circ} = \frac{80.0\text{J}}{10.0\text{g} \times 20.0^\circ C} = \boxed{0.4100\ J/g \cdot ^\circ C}$

18.) $M_{Total} = M_1 + M_2 + \ldots$

$M_1 =$ $N = 14.01\ g/mol$, $H_4 = 1.008\ g/mol \cdot 4 \} \times 2 = 36.084\ g/mol$

$M_2 =$ $C = 12.01\ g/mol$, $O_3 = 16.00\ g/mol \times 3 \} = 60.01\ g/mol$

$M_{Total} = 36.084\ g/mol + 60.01\ g/mol$

$= \boxed{96.09\ g/mol}$

19.) $100\,g$ $50\,g/mol$ moles = ?

$$\frac{100\,g}{50\,g} \left| \frac{1\ mol}{} \right. = \boxed{2\ moles}$$

scratch paper Name: Nadia Diaz ID: 09896716

#20 100g
× 50g
5,000 mol

scratch paper Name: Abel Figueroa ID: 0831812

H_2 / H = 3.024 =

$\frac{110.98\ g/mol}{131.011\ g/mol} \times 100 = 84.71\%$

(7) $CaCl_2 \cdot 2H_2O$

Percent of $CaCl_2$

Ca - 40.08

Cl - 35.45 x 2 = 70.9 = 110.98 g +

$2H_2 = 2.016 \times 2 = 4.032$

O - 15.999

20.031 g = 131.011 g

(19) 100 g total

50 g molar mass

$100\ g \times \frac{1\ mol}{50\ g} = 2.00\ mol$

$\frac{1\ mol}{molar\ mass}$

(9) 10.0 g Zn cooled 30°C to 10°C (F) Energy -80.0 J, what is C?

$C = \frac{q}{m \cdot \Delta T}$ $C = \frac{-80.0\ J}{10.0\ g \cdot (10-30)\ -20\ °C} = \frac{-80.0\ J}{-200\ g/°C} = 0.400\ J\ (g/°C)$

(8) 6.022×10^{23} atoms Na

$\frac{6.022 \times 10^{23}}{6.022 \times 10^{23}} = 1\ mol\ Na$

(18) $(NH_4)_2CO_3$

N - 14.007 x 2 - 28.014

H - 1.008 x 8 - 8.064

C - 12.011

O - 15.999 x 3 - 47.997

96.086 ≈ 96.09 g

(1) 4 - O = O has 6 valence x 4 = 24 ClO_4^-

Cl - 7 valence 24 + 7 = 31 + 1 = 32





scratch paper Name: Alexander Ruiz ID: 0913670

5. $CaCl_2 \cdot 2H_2O$ $\frac{(\overset{Ca}{40.08} + \overset{Cl_2}{70.9})}{147.01} \times 100 \approx 75.49$

Ca = 40.08 g/mol = 40.08

Cl = 35.45 g/mol x 2 = 70.9 = 147.01

H = 1.008 g/mol x 4 = 4.032

O = 15.999 g/mol x 2 = 31.998

6. 10.0 g = Zn = m

30.8 → 10.0 °C = ΔT

-80.0 J = q

Find C_p $C_p = \frac{q}{m\Delta T} = \frac{-80.0\ J}{10.0 \times (30^\circ C - 10.0)} = -0.400\ J/(g^\circ C)$

15. N = 14.01 H = 1.008 C = 12.01 O = 16.00

$(NH_4)_2CO_3$

C = 12.01
O x 3 = 48.00 → 60.01

N = 14.01
H x 4 = 4.032 = 18.042 x 2 =

36.084 + 60.01 = 96.094 g/mol

20. element = 100 g

molar mass = 50 g/mol $\frac{100\ g}{50\ g/mol} = 2$ mole

How many moles

21. Polar = $\Delta E > 0.5$ C-H = 0.4 X non
Nonpolar = $\Delta E < 0.5$ O-H = 1.4 ✓ Polar
Ionic $\Delta E > 2.0$ Na-Cl = 2.1 × Ionic
C-C = 0 × non

25. 6.022×10^{23} = 1 mol = 1 mole of Na

34

1.0 mol of SO_2 = 32.06 + (15.999)×2 64.058 × 1 = 64.058

0.5 mol of CO_2 = 12.011 + (15.999) × 2 44.009 × 0.5 = 22.0045

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+ 2+ 3+ 4+ 3- 2- -

$\Delta T = 10.0°C - 30.0°C = -20.0°C$

(7) $2(1.008) + 15.999 = 18.015$

$\times 2$

$40.08 + 2(35.45) = 36.03$

$= 110.98$

$+ 36.03$

147.01

$\frac{36.03}{147.01} \times 100 = 24.51\%$ of water

$100 - 24.51 = 75.49\%$ of $CaCl_2$

(8) 6.022×10^{23} of Na

$mol = 6.022 \times 10^{23}$

1 mol of Na ?

Other way?

6.022×10^{23} of NaCl $mol / 6.022 \times 10^{23}$

$6.022 \times 10^{23} \times 1$

$= 6.022 \times 10^{23}$

$= 1$ mol

(9) $c = q / m\Delta T$

$m = 10.0\ g$

$q = -80.0\ J$

$c = \frac{-80.0}{10.0(-20.0)}$

35 g f.s

$c = 0.4 \rightarrow 0.400\ J/g°C$

(18) $N = 2 \times 14.01 = 28.02$

$H = 8 \times 1.008 = 8.064$

$C = 1 \times 12.01 = 12.01$

$O = 3 \times 16.00 = + 48$

96.094

$96.09\ g/mol$

(19) $\frac{100}{100/50} = \frac{100}{2} = 50$ mol ?

$mol = \frac{g}{g/mol}$

(1)



$Cl = 7, 7$

$O = 6 \times 4 = 24$

$+ 1$

32 electrons

$Mg_3(PO_4)_2$ Mg^{2+} PO_4^{3-}

H C O Cl

2.1 2.5 3.5 3.0

$\times 0.4$ 1.4

$3.6 - 0.9 = 2.1$

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(7) ClO_4

$7 + 4(6)$

$7 + 24 + 1$

$= 32$

mass = moles × molar mass

1 mole = 6.022×10^{23}

(7.) $CaCl_2 \cdot 2H_2O$

$CaCl_2 = 40.08 + 2(35.45) = 40.08 + 70.9 = 110.98$

$H_2O = 2(1.008) + 15.999 = 2.016 + 15.99 = 18.015$

Total = 147.01

$2H_2O = 2(18.015) = 36.03$

$\frac{110.98}{147.01} \times 100$

(9.)

$= 10 - 30$

$Q = mc\Delta T$

$q = \frac{Q}{m\Delta T} = \frac{+80.0}{10.0 \times (+20)}$

$\frac{80}{10 \times 20} = \frac{80}{200}$

(10.) mass = moles × molar mass

$moles = \frac{mass}{molar\ mass}$

$mole = \frac{100}{50} = 2$

20. C — H O — H Na — Cl C — C

2.5 — 2.1 3.5 — 2.1 0.4 — 3.0 2.5 — 2.5

0.4 1.4 2.1 0

3.5 / 2.1 / 1.4

2.5 / 2 / 0.4

3.0 / 3.0 / 0

C — H O — H $N^+ - Cl^-$ C = H

polar Hydroge ionic

C — C

non Polar

8. = $(NH_4)_2 CO_3$

$N_2 H_8 C O_3$

$2(14.01) + 8(1.008) + 12.01 + 3(16.00)$

$28.02 + 8.064 + 12.01 + 48 = 96.094$

ionic

Sumandeep kaur
110498

(6) ClO_4

$7 + 4(6)$

$7 + 24 + 1$

$= 32$

(7) $CaCl_2 \cdot 2H_2O$

$CaCl_2 = 40.08 + 2(35.45)$

$= 40.08 + 70.9 = 110.98$

$H_2O = 2(1.008) + 15.999$

$= 2.016 + 15.99 = 18.015$

$2H_2O = 2(18.015) = 36.03$

$Total = 110.98 + 36.03 = 147.01$

$= \frac{110.98}{147.01} \times 100$

(9) $Q = mc\Delta T$

$Q \quad c = \frac{Q}{m\Delta T} = \frac{-80.0}{10.0 \times (-20)}$

$= \frac{+80.0}{+200} = 0.4$

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$2 \cdot 1.008$

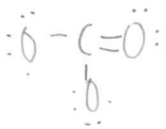
$2.016 + 15.99 = 18.006$

$\times 2$

36.012

$40.08 + (35.45 \cdot 2) = 110.98$

10.0g Zn Energy transfer −80J



(1)

O $6 - 4 = 24$

Cl 7

$24 + 7 = 31$

(18)

N $14.01 \cdot 2 = 28.02$ C 12.01

H $1.008 \cdot 8 = 8.064$ O $16.00 \cdot 3 = 48$

$48 + 28.02 + 12.01 + 8.064 = 96.094$

96.09 g/mol

(19)

100g 50g/mol

$100g \cdot \frac{1g}{50\ mol} = 2mol$

scratch paper Name: Marcella Perez ID: 1046135

Q5

$\frac{110.98\ g/mol\ CaCl_2}{147.012\ g/mol\ CaCl_2 \cdot 2H_2O} = .754904362 \cdot 100\% = \lceil 75.49\%\ CaCl_2 \rfloor$

$CaCl_2 = 40.08\ Ca + 2(35.45)\ Cl = 110.98\ g/mol\ CaCl_2$

$2\ H_2O$ (not water ??) $= 4(1.008\ H) + 2(16\ O) = 36.032\ g/mol\ 2H_2O$

$110.98 + 36.032 = 147.012\ g/mol\ CaCl_2 \cdot 2H_2O$

Q6 $10.0\ g\ Zn$ $30°C \rightarrow 10°C$ $-80.0\ J$

$q = -80.0\ J$

$c_p =$ find

$\Delta T = (-20°C)$

$m = 10.0\ g$

$-80.0\ J = c_p \times -20°C \times 10.0\ g$

$-80.0\ J = c_p \times -200\ g°C$

$\lceil c_p = .400\ J/(g°C) \rfloor$

Q15

$(NH_4)_2CO_3$

$NH_4 = 14.007\ N + 4(1.008\ H) = (18.039)(2) = 36.078\ g/mol\ (NH_4)_2$

$36.078\ (NH_4)_2 + 12.011\ C + 3(16)\ O = 96.089\ g/mol\ (NH_4)_2CO_3 \approx \lceil 96.09\ g/mol\ (NH_4)_2CO_3 \rfloor$

Q19

~~$Mg^{2+}\ PO_4^{3-}$~~

$Mg_3(PO_4)_2$

Q20

$100\ g \cdot \frac{1\ mol}{50\ g} = \lceil 2\ mol \rfloor$

Q25 6.022×10^{23} of Na

$Na = 22.99\ g/mol$

$22.99 \times (6.022 \times 10^{23}) \approx \lceil 23.99 \rfloor$

scratch paper Name: Francisca Villatoro ID: 0728589

(5) $CaCl_2$

$40.08 + 2(35.45) = 75.49\%$

10.0g 30.0°C → 10.0°C −80.0 J

(5) $(NH_4)_2(O_3$

14.01 g/mol + H 1.008 g/mol + C 12.01 g/mol + O 16.00 g/mol

14.01 + 1.0008 +

~~43.0208~~

(20) $100_g \div 50 g/mol = 2 mol$

(25) $6.022 \times 10^{23} = 6.022 \times 10^{23}$